3. Methods

Under this section, both the theoretical methods (e.g. the mechanisms of the model) and the practical methods (e.g. how the model was used in practice) are covered. Beyond that, there are discussions on the potential risks of the study, as well as the reliability of the most used sources in the background section.

3.1 Theoretical Method

Hereunder is described the theoretical method that constitutes the model, as well as the method for data selection and for the implementation of the model in a Python program.

3.1.1 Model for Price and Deviation Prediction

Much of the novelty of this paper lies within this section, and consists of a uniquely designed method of deviation prediction, which can be used to recommend a point in time for the trading to take place, and to better estimate the price of the stock at that point. In order to obtain a deviation-adjusted price prediction at the seemingly most accurate time coordinate in the future, the model operates by the following procedure:

(1) Provide N adjacent, minute-interval data points covering a desired time period of the stock in question. The N data points are divided into a desired number—P say—of shorter periods, which will henceforth be referred to as *short historical periods (SHPs)* (see Figure 2). An approximate Fourier series $x_{a,n}$ ($a \in \{1, 2, 3, \ldots, P\}$) with R frequency components is rendered for each of these SHPs. However, should $R > M = (N-1)/2$ or $R > N/2$ (the former for even N and the latter for odd N), the model uses the least of R and M, or R and $N/2$. Next, the Fourier series of each SHP is allowed to overlap the next SHP, one data point (i.e. minute) at a time. However, the first overlap must include two points within the subsequent SHP. The reason for this will become evident in step (5), when the deviation prediction method is described. The process of overlapping continues until the overlapping SHPs cover all points of their respective subsequent SHP. In Figures 3a-3f, the overlapping is visually explained with a simple, small-scale example.

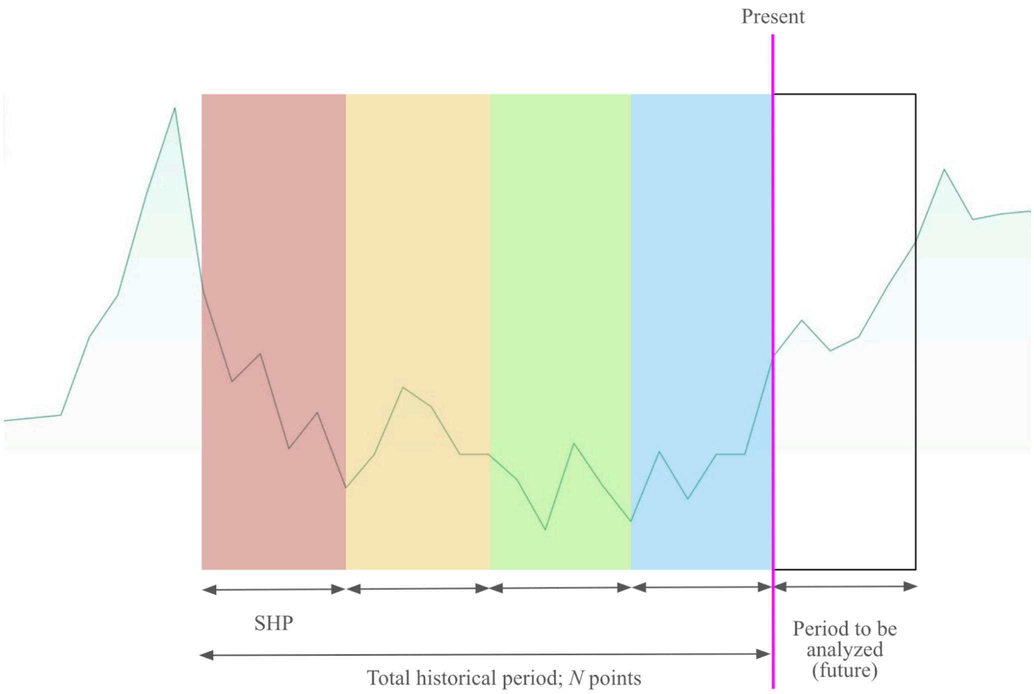

Figure 3a: Initial condition; overlapping has yet to take place. The graph in the background belongs to a randomly selected (and thus essentially arbitrary) stock. To the left of the present point are four adjacent SHPs, and to the right is the future period within which the model is allowed to make predictions. All SHPs, as well as the future period, are equally long.

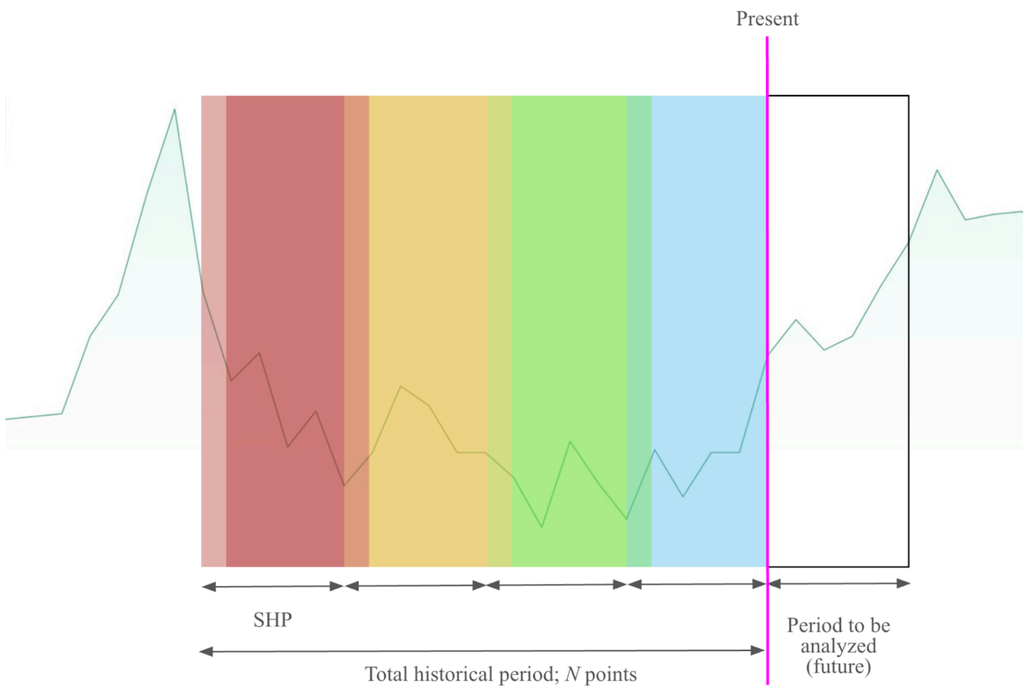

Figure 3b: First overlap step. All SHPs are allowed to overlap except the last, i.e. that which precedes the present point.

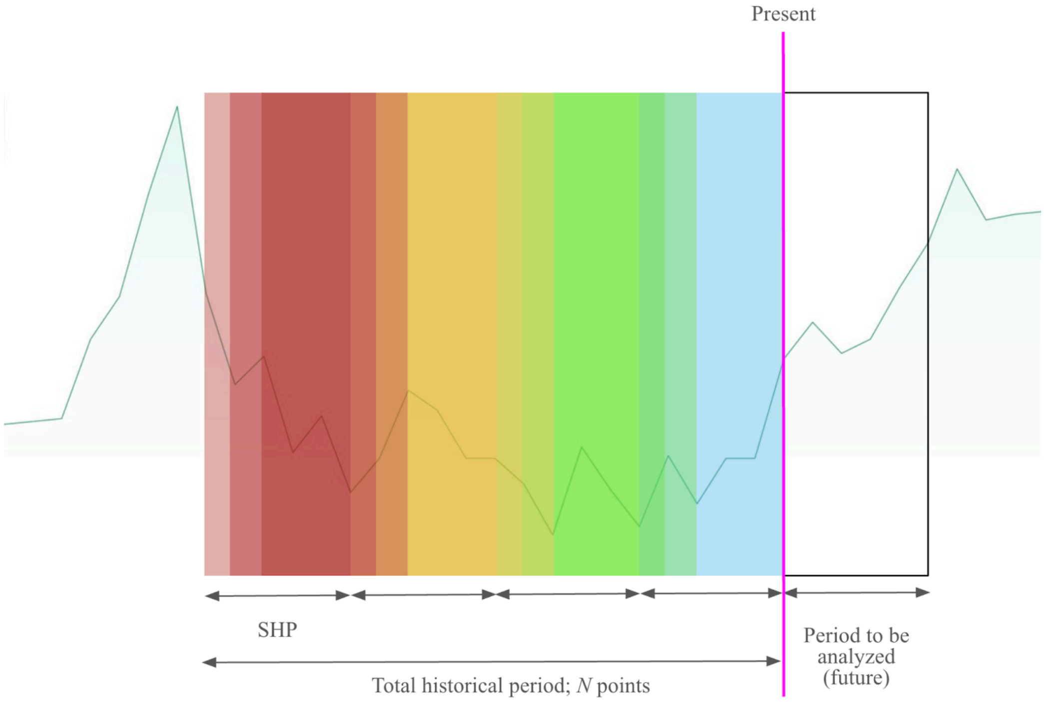

Figure 3c: Second overlap step

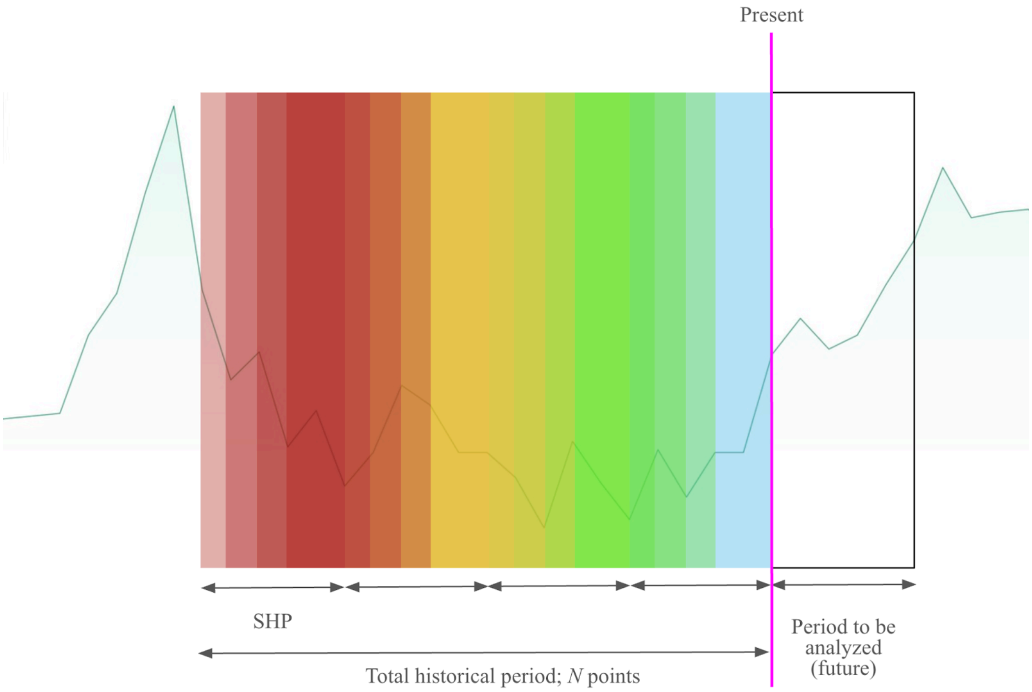

Figure 3d: Third overlap step

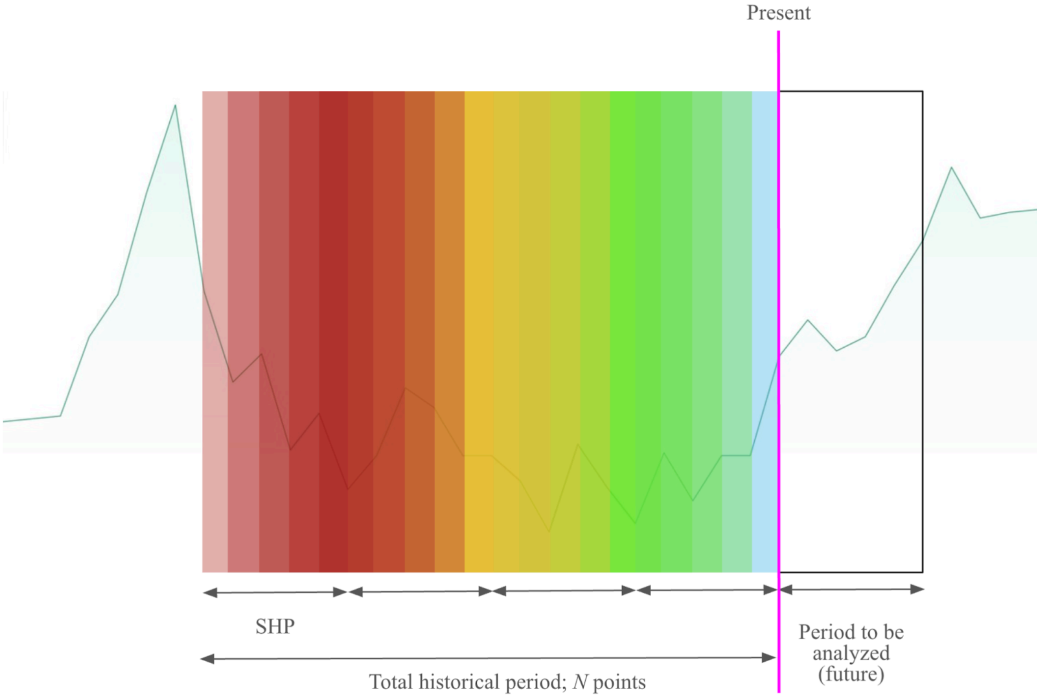

Figure 3e: Fourth overlap step

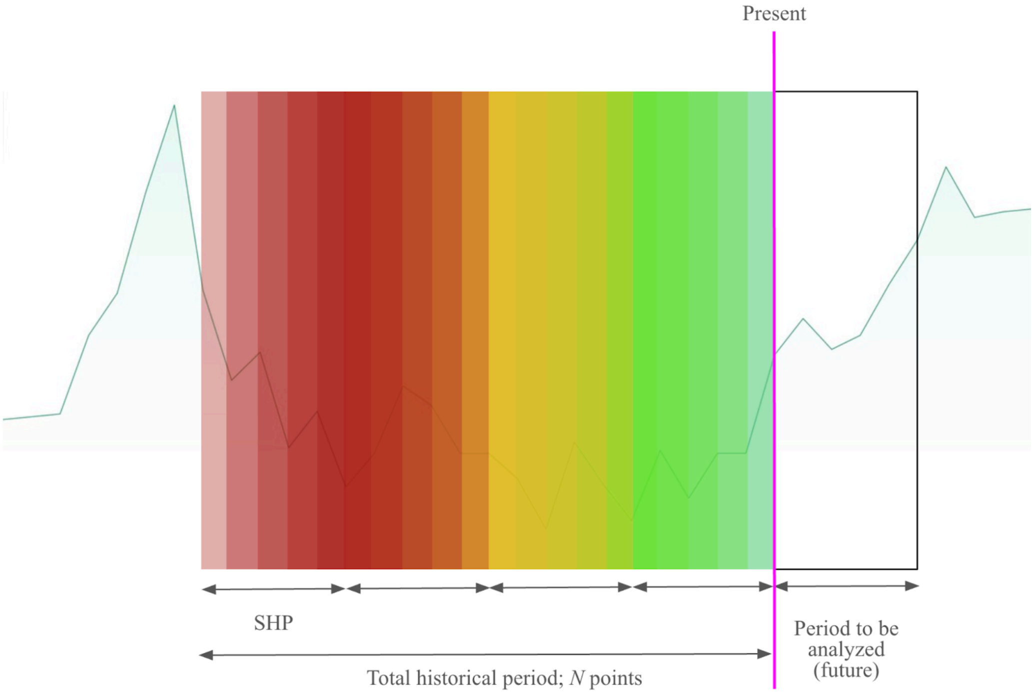

Figure 3f: Fifth overlap step; the overlap covers the entire subsequent SHP.

(2) Next, the function value of a Fourier series $x_{a,n}$—i.e. the stock price as predicted by the function—is computed for each *defined* point of the subsequent SHP. Here, *defined* means that the point is included in the overlap. For instance, in the second

overlap step, which includes three points within the next SHP, the function value will only be calculated at those three points; the fourth, fifth, etcetera, are outside the domain of $x_{a,n}$, and hence undefined.

This step is executed for all SHPs except the last—that which is adjacent to the present (the purple line in Figures 1-5)—and for all overlap steps. It should be clarified that the Fourier series of the last SHP, $x_{P,n}$, does not overlap, as there is no data, no subsequent SHP, to overlap (because the subsequent points do not yet exist; they are in the future).

It would be illogical to assume that the Fourier series can be used to predict prices for a greater amount of points in the future than the amount of historical points that has been used to generate it. This is why the maximum overlap is exactly one SHP, as mentioned in step (1). At this point, it is appropriate to define the number of points comprised within an SHP as Q, where $Q = \frac{N}{P}$ (given P SHPs and a total of N historical points used by the model).

(3) Every function value is then divided by the actual price at that point, yielding a quotient describing the precision of the Fourier series—a *precision quotient*. By means of subtracting 1 from the *precision quotient*, the latter is transformed into an expression of deviation: For quotients greater than 1, the deviations become positive, whereas quotients less than 1 result in negative deviations. For clarity in the subsequent step, the deviations will receive numbers according to which point they belong to: The deviation at the first point for which a function value has been obtained will be labeled as *Deviation 1*. Its equivalents at the second, third, and *a*th points at which function values have been computed are *Deviation 2*, *Deviation 3*, and *Deviation a*, respectively.

(4) Next, an average of the deviations is computed for each set of *corresponding* points, for example the fourth point of each SHP. The sets of corresponding points are specific to each overlap step: For instance, the set containing the fourth point of each SHP is not identical in the fourth, fifth, sixth, etcetera, overlap steps. In order to avoid that positive and negative deviations partially or completely nullify each other, absolute values are used for all deviations. Henceforth, the averages rendered in this step will be called *average deviations (ADs)* and denoted as δ. Provided, once again, that the number of SHPs is P:

$$\delta = \frac{\sum_{a=1}^{P-1} |Deviation\ a|}{P-1}.$$

(5) Subsequently, the AD of each set of corresponding points—including the sets from different overlap steps—is compared to the others; the set of corresponding points for which the AD is closest to 0 (this AD will be denoted as δ_{min}) constitutes the *recommended* set of points. This means that the operation to be performed in the future (buy, sell or short) should be executed at the *relative time coordinate (RTC)* of the recommended set of points. The RTC is defined as the difference in time (Δt) between the recommended point and the starting point (indexed with 0) of an SHP. Say the recommended set of points is constituted by the second point of each SHP, and that the points are found in the fifth overlap step. Then, the relative time coordinate is 2, and, accordingly, the investor is recommended to take action two points (minutes) into the future. Moreover, the actual prediction (that which is based on the last SHP and goes into the future) must have the correct overlap step—the fifth in this example.

Now, recall what was stipulated in step (2): The shortest allowed overlap step implies an overlap of two points onto the subsequent SHP. The reason is that the indexing of the points within the SHPs is zero-based, and this indexing is extended onto the present and future points. More specifically, the present point is regarded as a future point with index 0. As previously stated, the maximum number of future points considered by the model is Q. However, the present point is included in this, meaning that the actual number of future points is, at most, $Q - 1$. Since the model recommends an RTC which is then applied to the future points, the recommended RTC may not be 0: The latter would be equivalent to recommending the present point, but the model must always recommend a future point. Therefore, the minimum overlap must contain two points from the subsequent SHP—those indexed 0 and 1—or the recommended RTC would be out of bounds (if 0 were the only alternative).

(6) Finally, the following is obtained:

$$Recommended\ RTC = n,$$

$$(1-\delta_{min})\,x_n \leq Estimated\ price \leq (1+\delta_{min})\,x_n,$$

(26)

where $n \in \mathbb{Z}^+$ and $1 \leq n \leq Q$. x_n is the function value at the nth point in the future, and is adapted to the overlap step at which the recommended set of points was found.

3.1.2 Data Selection, Performance Evaluation and Visualization

Once again, this paper serves as a proof of principle. Therefore, the data set selected for the study was small, yet sufficient for conclusions to be drawn: The model was applied to ten almost arbitrarily chosen stocks from the S&P 500 index. By *almost* is meant that the selection of stocks was overseen in part, so as to ensure that not all companies were within the same sector, as this might result in unnecessary bias. Moreover, the maximum amount of historical data to which the model should be applied was set to 600 points, or 10 hours; $N \leq 600$. Then remained the issue of assigning values to P, Q, and R. These were all parameters of unknown, but potentially great, significance. As such, they were combined in a number of different ways, and the model was tested for all combinations. Prior to determining these combinations, the parameters each received a domain: P and Q were given the same domain, namely the set of all divisors of the number 60 (the amount of points in the time series created in one hour), except for 1 and 60. Accordingly, $P, Q \in \{2, 3, 4, 5, 6, 10, 12, 15, 20, 30\}$. Call this set Y_1, and pay particular attention to the fact that the greatest allowed value of Q was 30, meaning that the prediction might extend at most 30 minutes into the future. Since $Q = \frac{N}{P}$, $N = P \cdot Q$, which would imply that $4 \leq N \leq 900$. Since N was defined to be no greater than 600, P and Q might not simultaneously evaluate to 30. All other combinations of the two were nonetheless allowed. Finally, R was given the set $Y_2 = \{5, 10, 15, 20\}$.

Towers (2022) defines the Cartesian product of two sets A and B ($A \times B$) as the set of ordered pairs (a, b) for which the first element is a member of A, and the second element is a member of B:

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

It then follows that the cardinality (the number of elements) of the set $A \times B$ is equal to the product of the respective cardinalities of A and B (Towers, 2022):

$$|A \times B| = |A||B|,$$